

Weather Pattern Analysis using Power BI

Question 1 : How do temperature patterns vary across cities and seasons?

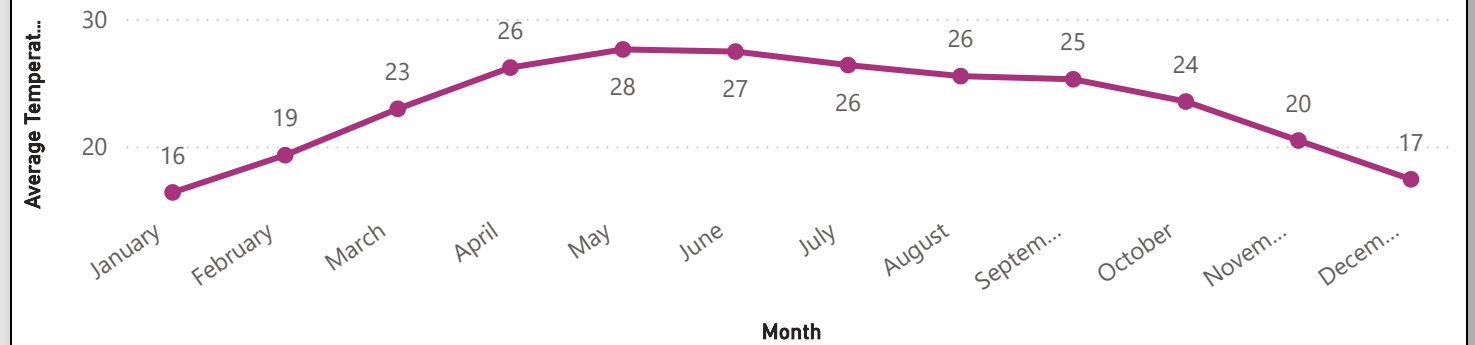
23.23

Average Temperature

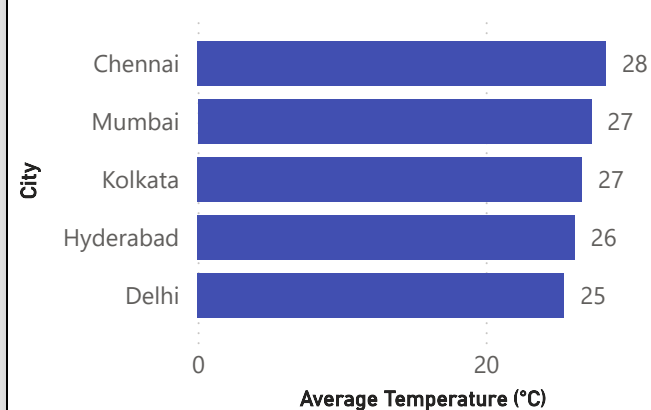
46.20

Maximum Temperature

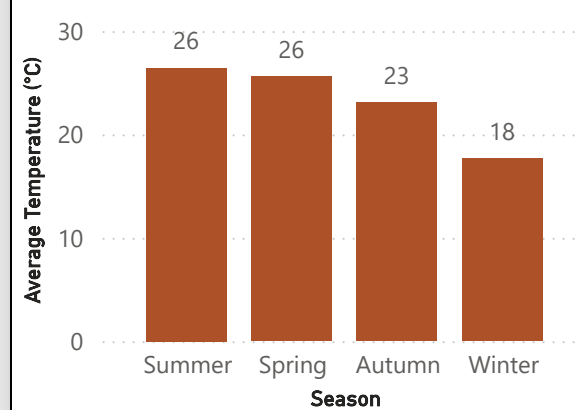
Monthly Temperature Trend



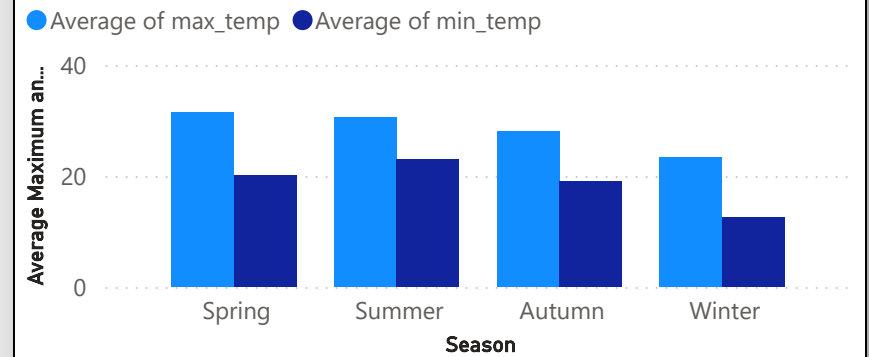
Average Temperature by city



Average Temperature by season



Average Maximum vs Minimum Temperature by Season



Question 2 : How do rainfall patterns vary across cities and seasons?

30.74K

Total Rainfall

4.21

Average Rainfall

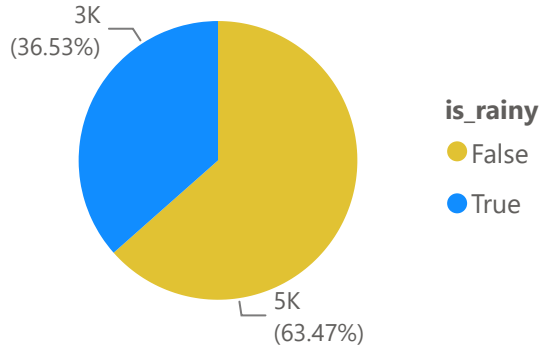
3K

Rainy Days

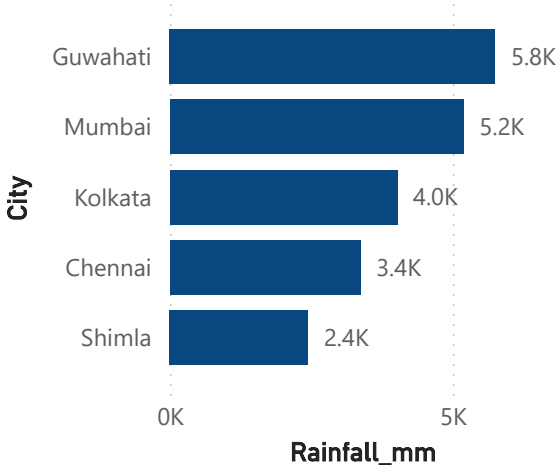
5K

Non Rainy Days

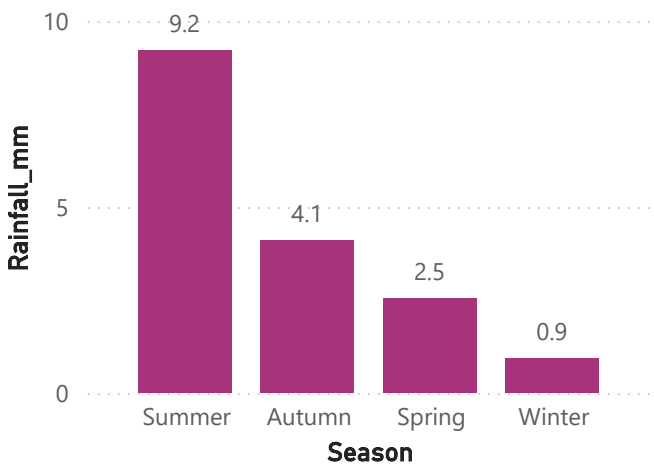
Rainy vs Non-Rainy Days



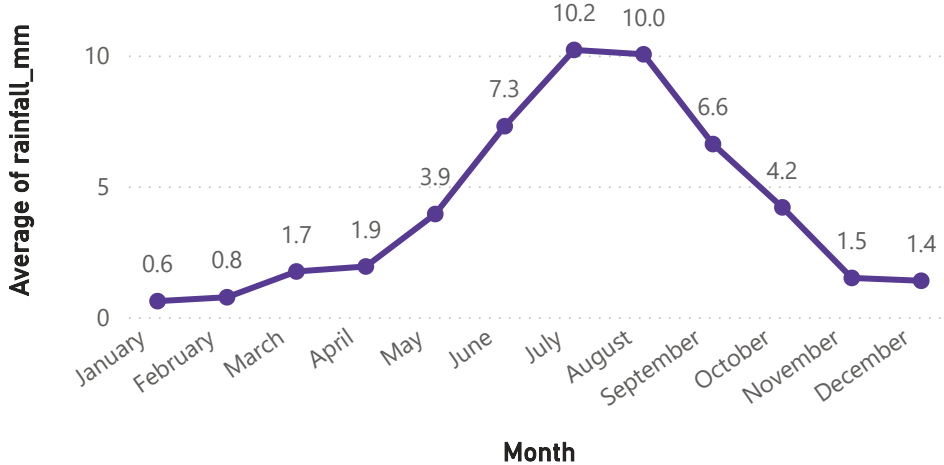
Total Rainfall by City



Average Rainfall by Season



Monthly Rainfall Trend



Question 3: Which cities experience the most extreme weather conditions?

46.20

Maximum Temperature

-14.90

Minimum Temperature

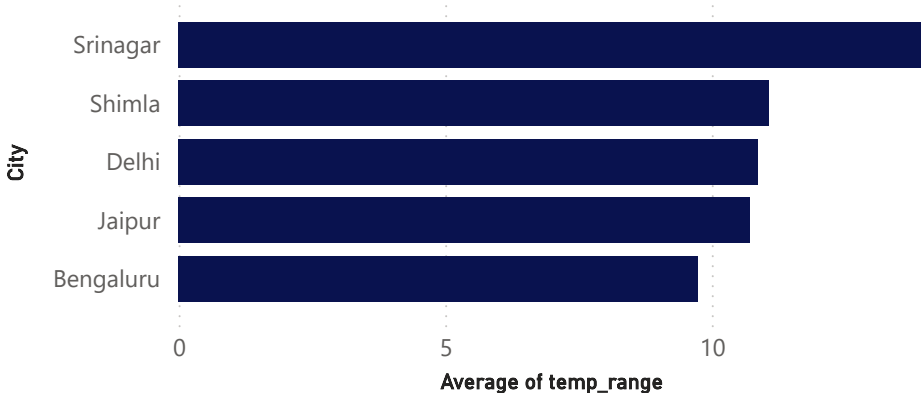
9.66

Average of temp_range

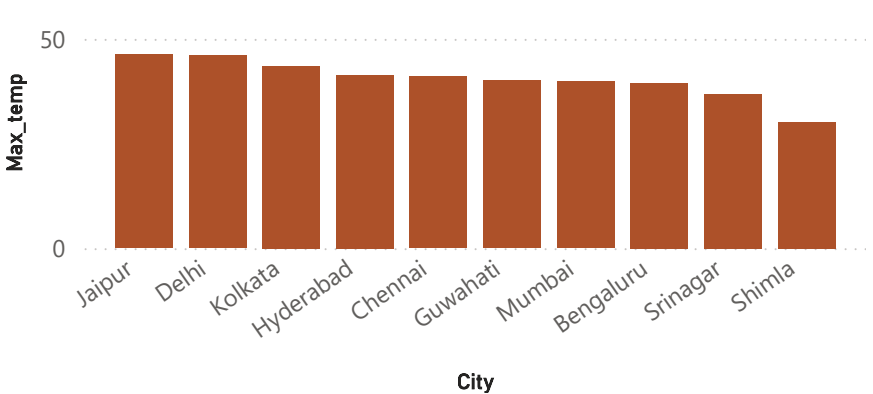
4.21

Average of rainfall_mm

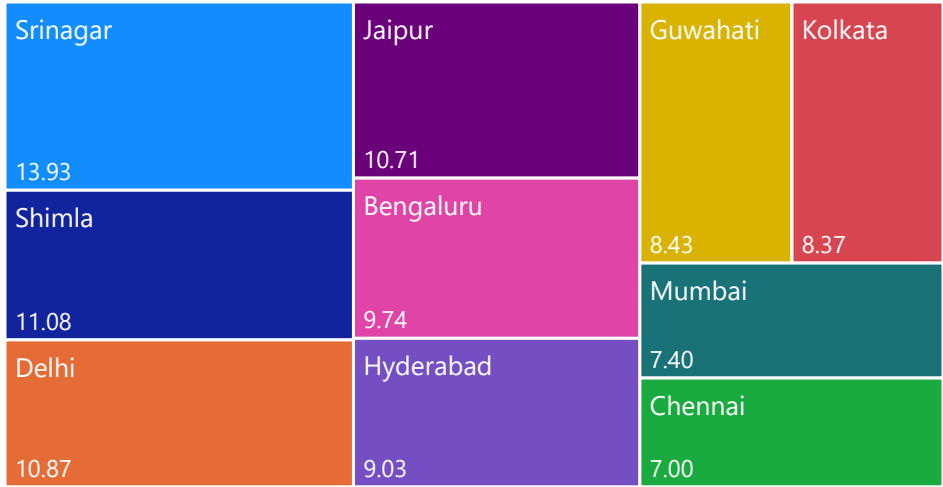
Average Temperature Range by City



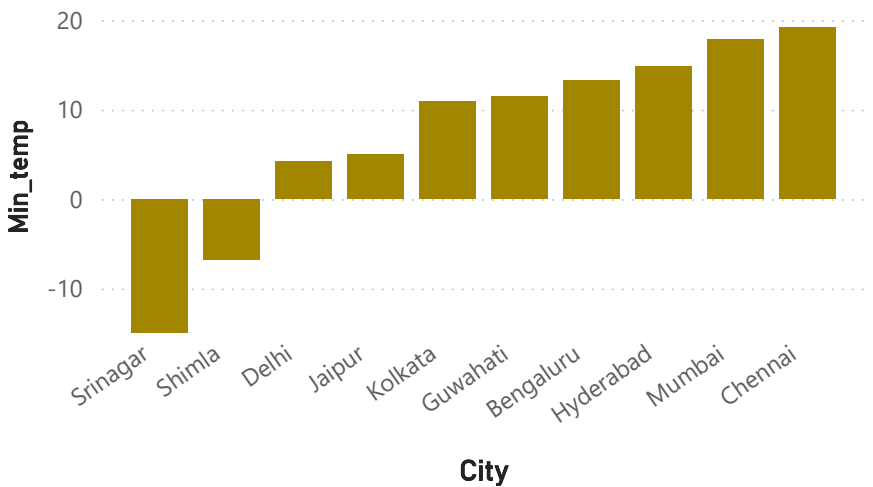
Highest Temperature Recorded by City



Average Temperature Range by City



Lowest Temperature Recorded by City



Question 4 : How have weather patterns changed from 2024 to 2026?

23.23

Average Temperature

46.20

Maximum Temperature

4.21

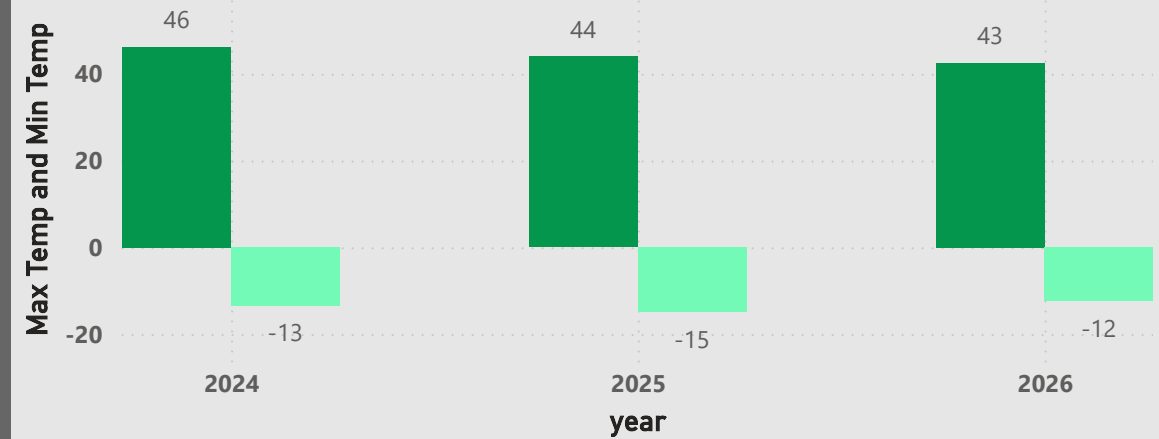
Average Rainfall

-14.90

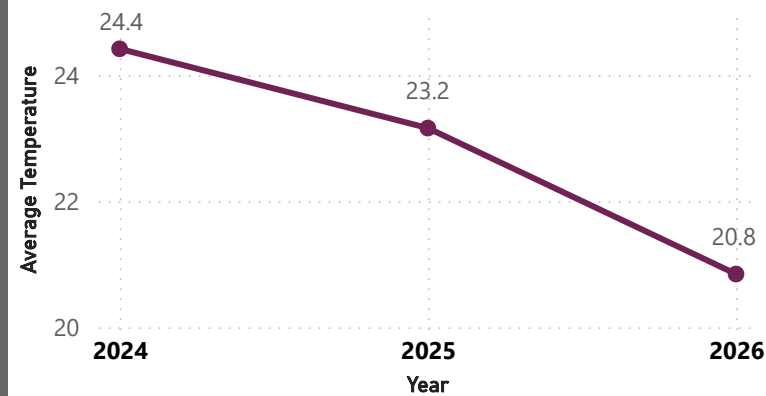
Minimum Temperature

Maximum vs Minimum Temperature by Year

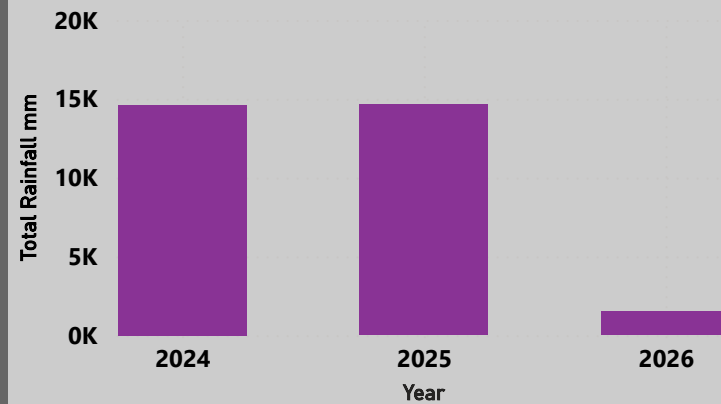
● Max of max_temp ● Min of min_temp



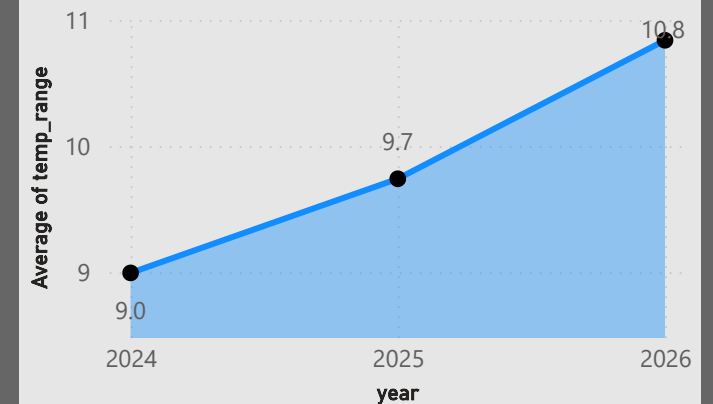
Average Temperature Trend (2024–2026)



Total Rainfall by Year



Average Temperature Range by Year



Question 5 : What are the key weather insights and overall trends from the dataset?

23.23
Average Temperature

30.74K
Total Rainfall

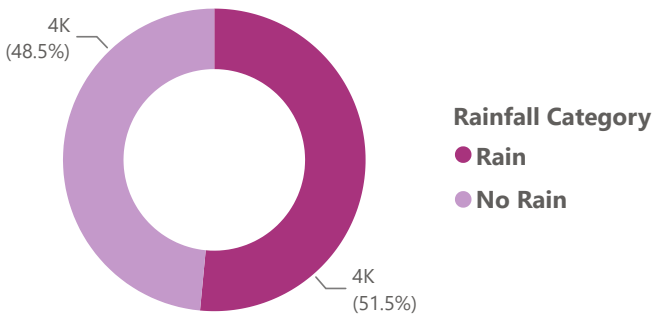
46.20
Maximum Temperature

-14.90
Minimum Temperature

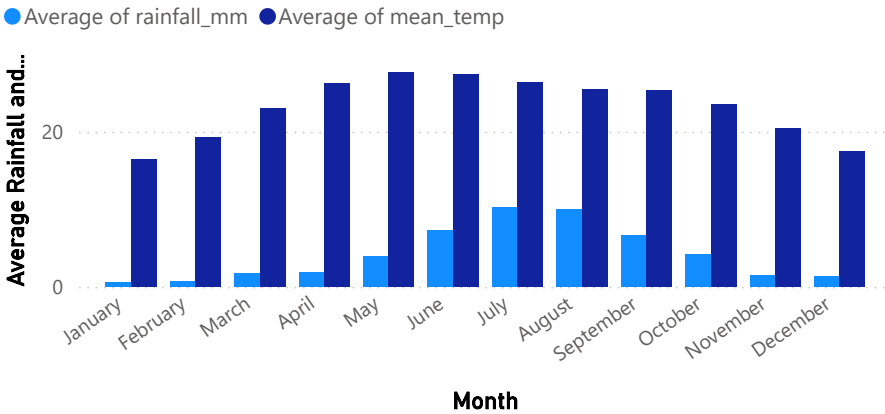
3K
Rainy Days

7K
Total Records

Distribution of Rainfall Categories



Monthly Temperature & Rainfall Trend



Seasonal Contribution to Total Rainfall

